

BookStore Management System

Project Title: BookStore MERN Application

College Name: Aditya College of Engineering and Technology

Department: Department of Computer Science & Engineering

Course: Bachelor of Technology

Technology: Full Stack Web Development (MERN)

Repository: <https://github.com/HemaSana21/bookstore>

S.No	Team Member	Role	Responsibilities
1	Sana Hema Naga Satya Bhavani	Team Lead	Project planning, team coordination, backend integration, final project management
2	Bala Sai Siraparapu	Full Stack Developer	Frontend development, backend API integration, database connectivity, testing and documentation
3	Appanapalli Sunitha	Frontend Developer	Designing React components, creating UI pages, improving user experience
4	AddankiVeeraVenkata SaiGanesh	Backend Developer	Developing REST APIs, server-side logic, authentication implementation
5	Sarup Bormon	Database & Testing Member	MongoDB database management, API testing, bug fixing and project support

1. Introduction

BookStore is a web-based e-commerce application developed using the MERN stack (MongoDB, Express.js, React.js, and Node.js). The application provides an online platform where users can browse, search, and purchase books conveniently. It supports three roles: User, Seller, and Admin. JWT authentication ensures secure login and role-based access

control. The application is designed with a responsive interface that works on desktops, tablets, and smartphones.

The platform allows customers to discover books by category, author, or genre, maintain wishlists, manage shopping carts, place orders, and review purchased books. Sellers can upload and manage books, monitor inventory, and fulfill customer orders. Administrators oversee the entire platform by managing users, sellers, books, and reports.

2. Problem Statement

Traditional bookstores rely on physical visits and manual inventory management, making book purchasing less convenient and prone to stock and record errors. A centralized digital platform addresses these issues by automating inventory, orders, authentication, and reporting.

This module was designed following software engineering best practices with emphasis on usability, security, scalability, maintainability, and responsive performance. Appropriate validation, error handling, and modular code organization ensure that the application is reliable and easy to extend in future releases.

3. Objectives

- Develop a secure online bookstore.
- Provide authentication using JWT.
- Allow searching, filtering, and purchasing books.
- Manage inventory and orders.
- Provide dashboards for User, Seller, and Admin.
- Maintain secure data in MongoDB Atlas.
- Deliver a responsive and scalable application.

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4. Scope

The application supports online book browsing, ordering, inventory management, seller operations, and administrative monitoring. It can be extended with payment gateways, AI recommendations, multilingual support, and mobile applications.

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5. Existing System

Existing systems often depend on physical stores and manual record keeping. They provide limited accessibility, slower service, and inefficient inventory management. Customers cannot easily compare books or track orders online.

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6. Proposed System

The proposed BookStore automates the complete purchasing workflow through a secure web application. It uses REST APIs, MVC architecture, JWT authentication, and MongoDB Atlas to provide reliability, scalability, and maintainability.

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7. Software Requirements

Operating System: Windows 10/11

Frontend: React.js

Backend: Node.js, Express.js

Database: MongoDB Atlas

IDE: Visual Studio Code

Testing: Postman

Version Control: Git & GitHub

Browser: Google Chrome

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8. Hardware Requirements

Processor: Intel Core i5 or above

RAM: 8 GB or above

Storage: 20 GB free

Internet Connection Required

Keyboard and Mouse

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9. Technology Stack

Frontend: React.js, HTML5, CSS3, JavaScript, Axios

Backend: Node.js, Express.js

Database: MongoDB Atlas

Authentication: JWT & bcrypt

Tools: VS Code, GitHub, Postman

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10. System Architecture

Presentation Layer: React user interface.

Application Layer: Express REST APIs, Controllers, Authentication.

Database Layer: MongoDB collections for Users, Sellers, Books, Orders, Reviews, Inventory.

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11. Database Design

Collections include Users, Sellers, Admins, Books, Orders, Reviews, Wishlist, Cart, and Inventory. Relationships ensure proper association between users, books, sellers, and orders.

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12. Frontend Implementation

Includes Home, Login, Registration, Product Listing, Book Details, Cart, Wishlist, Checkout, User Dashboard, Seller Dashboard, and Admin Dashboard. Reusable components improve maintainability.

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13. Backend Implementation

Uses MVC architecture with Models, Controllers, Routes, Middleware, and Config. JWT middleware secures protected APIs while controllers handle business logic and models manage MongoDB data.

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14. API Endpoints

Authentication APIs, Book CRUD APIs, Cart APIs, Wishlist APIs, Order APIs, Seller APIs, and Admin APIs following REST standards.

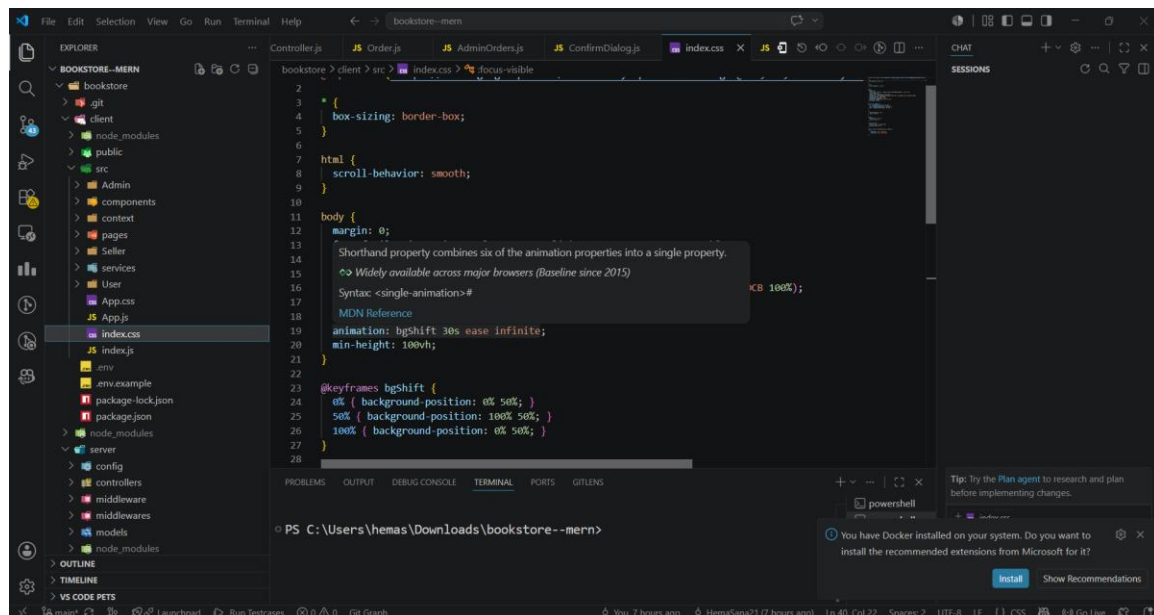
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15. Folder Structure

Frontend: components, pages, admin, seller, user, services, context.

Backend: config, controllers, middleware, models, routes, uploads, utils.

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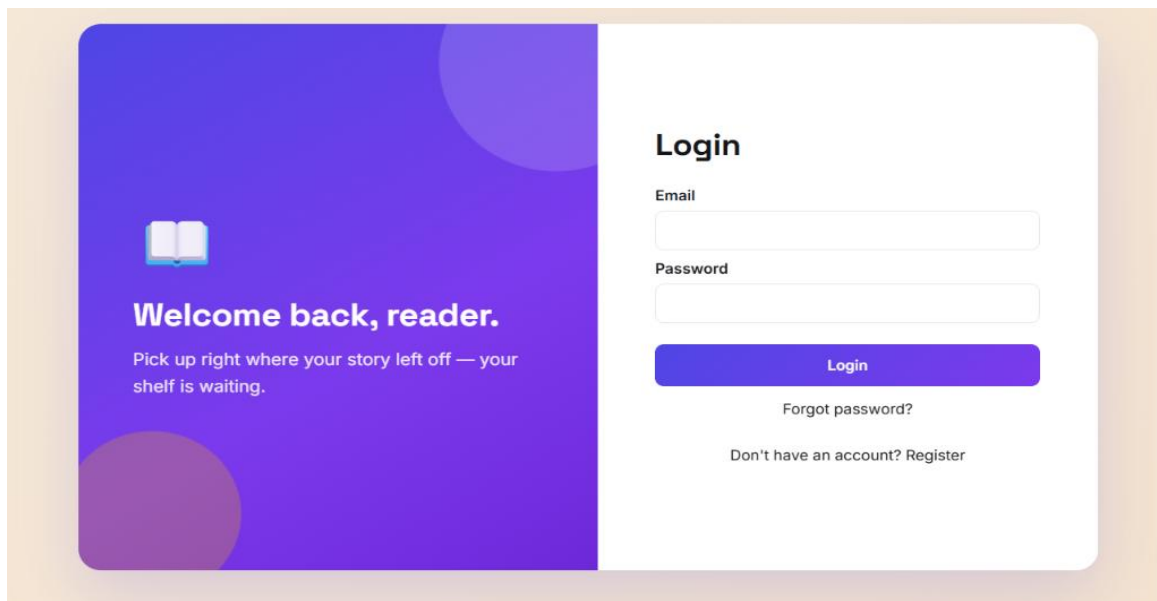
16. Testing

Functional testing confirmed successful registration, login, CRUD operations, cart management, order placement, seller management, and admin operations.

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17. Output Screenshots

- **Login Page**



2.RegisterPage



Join the shelf.

Create your account and start collecting stories worth keeping.

Create an Account

Full Name

Email

Mobile Number

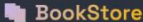
Password

Confirm Password

Register

Already have an account? [Login](#)

3.BookListPage



[Books](#)
[Login](#)
[Register](#)

Filters

Search

Category

All Categories

Language

All Languages

Min Rating

Any

Price Range

Sort By

Latest

All Books

-10%

Becoming Code 50

Malcolm Gladwell
SCIENCE

Becoming Code 50

Malcolm Gladwell

★ 0.0 (0)

₹655.2 ₹728

Add to Cart

-15%

Sapiens: Habits 49

James Clear
HISTORY

Sapiens: Habits 49

James Clear

★ 5.0 (1)

₹158.95 ₹187

Add to Cart

-10%

Clean A Memoir 48

Ryan Holiday
FICTION

Clean A Memoir 48

Ryan Holiday

★ 2.0 (1)

₹784

Add to Cart

-10%

The Power of Slow 47

James Clear
BUSINESS

The Power of Slow 47

James Clear

★ 5.0 (1)

₹450 ₹500

Add to Cart

4.Cart Page

Shopping Cart



Becoming Code 50
Malcolm Gladwell
- 1 +
[Delete](#)

₹655.2

Price Summary

Subtotal	₹655.2
Discount	-₹0
GST (5%)	₹32.76
Delivery	Free
Grand Total	₹687.96

5.Profile Page



Test User 1

user1@example.com

Personal Details

Email (cannot be changed)

Full Name

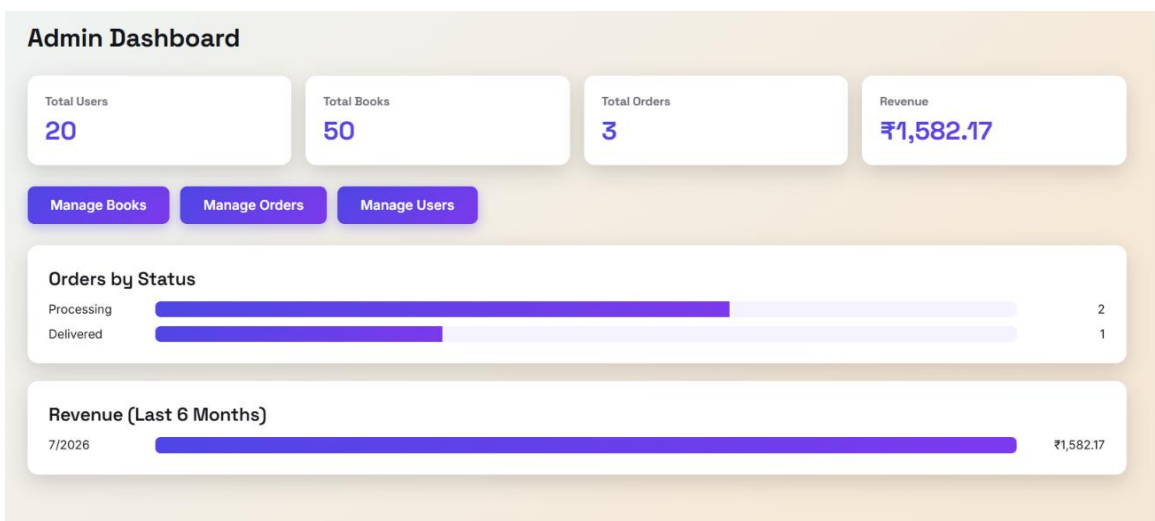
Phone

Shipping Address

Address Line 1

City / State / Pincode

6.Admin Dashboard



18. Future Enhancements

Payment gateway integration, AI recommendations, push notifications, mobile application, multilingual support, analytics dashboard, coupon system, and live order tracking.

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18. Conclusion

BookStore demonstrates a complete MERN-based e-commerce solution with secure authentication, responsive UI, role-based authorization, MVC architecture, RESTful APIs, and MongoDB integration. The project is scalable, maintainable, and suitable for real-world deployment as well as academic evaluation.

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